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Shale softening induced by CO_2 injection in the absence and presence of water: An experimental investigation based on nanoindentation

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ABSTRACT

Injecting CO_2 into shale gas formations to improve recovery and sequestrate CO_2 is regarded as a promising technology. Understanding the softening behavior of shale induced by CO_2 injection is crucial for the engineering decision of shale gas development and CO_2 geo-sequestration. In this study, the softening behavior of shale induced by CO_2 injection was investigated based on nanoindentation. Emphasis was placed on analyzing the differences in shale softening behavior in the absence and presence of water. CO_2 injection caused alteration of shale mineral composition and mechanical properties. The contents of quartz and muscovite increased after treatment, while albite, anorthite, and kaolinite decreased. The nanoindentation results were analyzed using the Gaussian mixture model and upscaled using Mori-Tanaka model. Three mechanical phases (i.e. soft phase, intermediate phase, and stiff phase) were identified through Gaussian mixture model. Under the condition of CO_2 treatment, both the hard and soft phases softened, while the intermediate phase did not. Under the condition of CO_2+H_2O treatment, all three phases - hard, intermediate, and soft - experienced significant softening. The macroscale elastic modulus of shale sample was estimated through Mori-Tanaka method. Compared with untreated sample, the decrease in macroscale elastic modulus of CO_2 treated sample is 10.92 %, and that of CO_2+H_2O treated sample is 45.76 %. The results of this study show that shale softening induced by CO_2 injection is more severe in the presence of water than that in the absence of water.

1. Introduction

The successful development of shale gas in the United States has transformed the global energy landscape [1]. As an alternative to conventional natural gas, shale gas has garnered significant interest from other major energy-consuming nations including China [2] and Canada [3]. Owing to the inherent low porosity and permeability of reservoirs, the implementation of reservoir stimulation strategies is typically requisite for the attainment of commercial exploitation [4]. The technical system coupling horizontal drilling with multi-stage hydraulic fracturing is currently the predominant technique employed for shale gas reservoir stimulation [5]. Hydraulic fracturing is the process of injecting fluid into a target formation under sufficiently high pressure to fracture the rock. The purpose is to enhance permeability, thereby increasing production [6]. However, hydraulic fracturing technique based on water-based fracturing fluids may bring some negative impacts, such as substantial water consumption [7], potential reservoir damage [8], and environmental threats from fluid flowback [9]. Consequently, there is an urgent need for more advanced stimulation technologies to alleviate these issues.

Injecting CO_2 into deep shale gas formations to improve recovery is regarded as a promising technical solution [10]. When both temperature and pressure exceed 31.1 °C and 7.38 MPa respectively, CO_2 enters a supercritical state. Due to its low viscosity, high density, and elevated diffusivity, supercritical CO_2 (Sc-CO2) is considered to be an ideal waterless fracturing fluid for shale gas development [11]. Compared with hydraulic fracturing, using Sc-CO_2 as a fracturing fluid can lower the breakdown pressure of shale reservoirs and generate more complex and conductive fracture networks [12]. Zhang et al. conducted physical simulation experiments and found that using Sc-CO_2 as a fracturing fluid can reduce the pressure required to initiate fractures by more than 50 % compared to hydraulic fracturing [13]. Besides, due to the stronger

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affinity of shale for CO_2 than for CH_4, the adsorbed CH_4 can be replaced by injected CO_2 through competitive adsorption [14]. CO_2/CH_4 competitive adsorption can increase the shale gas recovery rate while achieving CO_2 geological storage.

Nonetheless, CO_2 injection can alter the physical and chemical properties of shale. Yin et al. reported the alteration of shale mineral composition and pore structure after exposure to Sc-CO_2 [15]. The research by Lu et al. indicated that injecting CO_2 into shale reservoirs can reduce the water-wettability of the shale, thereby increasing the shale gas recovery [16]. After exposure to CO_2, the mechanical properties of shale may deteriorate, i.e. shale softening. Shale softening induced by CO_2 injection will have important impacts on shale gas development and CO_2 geo-sequestration, such as increasing proppant embedment depth, decreasing fracture conductivity, fracability [17], and caprock integrity [18]. Numerous investigations have been conducted to characterize the softening behavior of shale after CO_2 exposure, and it was found that the uniaxial compressive strength (UCS), elastic modulus (E) [5], triaxial compressive strength (TCS) [19], tensile strength (TS) [20], and fracture toughness (K_{IC}) [21] of shale decreased after CO_2 exposure. Moreover, the effect of CO_2 on shale softening could be more significant in the presence of water [18]. Lyu et al. found after a 10-day treatment with CO_2, the UCS of shale slightly decreased by 4.62 %, but the value for CO_2+H_2O treatment declined by 38.87 % [22]. Zhou et al. investigated the mechanical properties alteration of shale exposed to different fluids and found that the reduction of UCS for water, subcritical CO_2, subcritical CO_2+water, Sc-CO_2, and Sc-CO_2+water treated shale samples are 2.72 %, 5.45 %, 10.12 %, 12.06 %, and 21.01 %, respectively. The elastic modulus reductions for water, subcritical CO_2, subcritical CO_2+water, Sc-CO_2, and Sc-CO_2+water treated shale samples are 2.49 %, 4.83 %, 6.63 %, 10.03 %, and 12.38 %, respectively [23]. Compared with the brine and corresponding single CO_2 conditions, further mechanical weakening in shale was observed in the Sub/-or Sc-CO_2+brine soak conditions [24].

It should be mentioned that most previous studies used macro-mechanical testing methods, such as uniaxial compression, triaxial compression, and Brazilian splitting, to investigate shale softening induced by CO_2. However, shale is a multiphase heterogeneous material. It is difficult to fully characterize the softening behavior of shale by macromechanical testing methods solely [25]. Microscale mechanical investigation can positively supplement the understanding of shale softening behavior. Nanoindentation has been proven to be a powerful technique for characterizing the micromechanical properties of shale, which can capture the mechanical response at a submicron scale. The review paper by Ma et al. comprehensively summarized recent applications of nanoindentation technology in rocks [26]. Nanoindentation makes it possible to characterize the mechanical behavior of individual phases within the shale. Ulm et al. developed a test technique to identify different mechanical phases in shale based on grid nanoindentation and statistical deconvolution [27]. Liu et al. successfully applied it to the estimation of macro-mechanical properties of Bakken shale [28]. Another advantage of nanoindentation is that it only requires a small volume of sample for testing. This breaks the limitation of macro-mechanical testing methods, which requires standard specimens, making it possible to characterize the mechanical properties of fragmented rock cores from underground. Several studies have been conducted to investigate the softening behavior of shale at the microscale. Liu et al. studied the softening behavior of Marcellus shale after water vapor treatment through nanoindentation. They found that the elastic modulus of shale exhibited a reduction of 0.6–15 % after 20 days of water treatment, while its hardness decreased by 8.6–17.8 % [29]. Shi et al. characterized the micromechanical property alteration of shale after Sc-CO_2 treatment and found that the decrease of Young's modulus and hardness was more significant in clay-rich areas than in quartz-rich areas [30]. Cheng et al. conducted a three-month CO_2-water-shale interaction experiment and investigated the variation of the micro-mechanical properties of shale using nanoindentation. They found that after CO_2-water treatment, different shale minerals experienced different reductions of elastic modulus and hardness because of the dissolution induced the alterations in crystal structures [31]. Ozotta et al. measured the changes in micromechanical parameters of Middle Bakken shale under different Sc-CO_2 treatment durations, and upscale the micromechanical parameters to macroscale through homogenization methods [32]. Li et al. compared the microscale softening behaviors of Shahejie shale induced by water treatment and CO_2 treatment. The research findings suggest that the degree of softening in shale after CO_2 soaking is less than that observed in shale after water soaking [33]. To date, the comparative studies on shale softening induced by CO_2 in the absence and presence of water are still limited. The research gap is as follows: At the macroscale, prior studies can only obtain the alteration of the bulk mechanical properties of shale, but cannot determine the softening of its constituent components. At the microscale, although there are some studies on shale softening under single treatment conditions of CO_2 or CO_2+H_2O, comparative studies under the same sample and experimental conditions are still lacking.

It is significant to be aware of the softening behavior of shale under the treatment of CO_2, especially the difference in softening degree in the absence and presence of water, since this is related to the engineering decision of shale gas development and CO_2 geo-sequestration. In this study, shale samples from Yanchang Formation were treated with CO_2/CO_2+H_2O. The micromechanical properties of shale under different treatment conditions were measured using nanoindentation. Based on the statistical deconvolution method, the softening behaviors of different components in shale were analyzed, and the alterations in the bulk mechanical properties of shale were obtained through homogenization method. The differences in softening behavior of shale under CO_2/CO_2+H_2O treatment conditions were compared. In addition, the alteration of shale minerals was characterized by X-ray diffraction (XRD). Finally, the softening mechanism of shale after CO_2 treatment in the absence and presence of water and its impact on related engineering issues was discussed. The innovation of this paper lies in the comparative analysis of shale softening behavior induced by CO_2 injection at the microscale, based on the consistent sample and experimental conditions. The softening degree of different mechanical phases in the absence and presence of water is determined under comparable conditions. Furthermore, the controlling role of the soft phase in the softening behavior of the Yanchang shale is revealed. The findings from our research can inform optimal strategies and practices for CO_2 fracturing and geological sequestration.

2. Materials and methods

2.1. Sample description and preparation

The shale sample is from the shale gas reservoir of the Triassic Yanchang Formation in the Ordos Basin, China, which is a typical continental shale gas reservoir in China. The core was cut into 10 mm × 10 mm × 10 mm sized cubes using a wire cutter. Precision polishing of the nanoindentation test surface is necessary. Firstly, different grades of sandpaper (800–5000 mesh) were used to polish the shale cubes, followed by additional polishing using 1- and 0.5-μm diamond slurry. Next, a Leica EM-RES 102 ion thinning instrument was employed for argon ion polishing. The shale samples were polished for 4 h at an accelerating voltage of 4 kV and polishing inclination of 3°. It should be mentioned that all nanoindentation tests were conducted on planes parallel to the bedding planes. A portion of the samples were crushed into 200 mesh powders for XRD analysis.

2.2. CO_2 treatment procedure

Two high-temperature and high-pressure reaction vessels produced by Shanghai Baikal Technology Co., Ltd. were used for CO_2 treatment procedure. The reaction vessels were made of corrosion-resistant

Hastelloy and were equipped with temperature and pressure monitoring modules as well as an automatic temperature control module. Vessel #1 was used for the CO_2 treatment experiment. After the shale samples were loaded into the reaction vessel, CO_2 was injected into the reaction vessel using a high-precision plunger pump (ISCO 260D, Teledyne Co., Ltd., USA) until the pressure stabilized at the set value. Vessel #2 was used for the CO_2+H_2O treatment experiment. Shale samples and 100 ml of deionized water were loaded simultaneously into the reaction vessel (the samples were completely submerged in the deionized water), after which CO_2 was injected into the reaction vessel until it reached the set pressure. According to the geological conditions, the treatment temperature and pressure were set at 50 °C and 12 MPa, respectively. The samples were taken out after 60-day treatment.

2.3. XRD analysis

Powder samples were tested at a rate of 5°/min ranging from 5° to 90° by an X-ray diffractometer (Bruker D8 Advance, Germany) to determine the mineral contents of shale before and after CO_2/CO_2+H_2O treatment (see Fig. 1).

2.4. Nanoindentation

2.4.1. Testing principles

Nanoindentation involves vertically intruding a diamond indenter with specific shape and material parameters into the sample surface, and withdrawing it after a set depth or load has been reached. A typical load-displacement curve is shown in Fig. 2. The hardness and elastic modulus of the sample can be calculated from the collected load-displacement curve [34]:

$$H = \frac{F_{\max}}{A_c} \tag{1a}$$

$$E = \left(1-\nu^2\right)\left[\frac{1}{E_r} - \frac{1-\nu_i^2}{E_i}\right]^{-1} \tag{1b}$$

where H is the hardness, $F_{\max}$ is the peak load, and A_c is the contact area. E and E_i are elastic moduli of the sample and indenter, respectively. ν and ν_i are the Poisson's ratios of the sample and indenter, respectively. $E_i = 1141$ GPa, $\nu_i = 0.07$. The value of ν is assumed to be 0.3 [35]. E_r is the reduced modulus, and is given by:

$$E_r = \frac{\sqrt{\pi}S}{2\eta\sqrt{A_c}} \tag{2}$$

where S is the contact stiffness, which is calculated by the initial stage of the unloading curve, $S = (dF/dh)_{h=h_{\max}}$. η is a constant related to the indenter geometry, and for a Berkovich indenter, $\eta = 1.034$ [36]. A more detailed calculation procedure can be found in the prior literature [26].

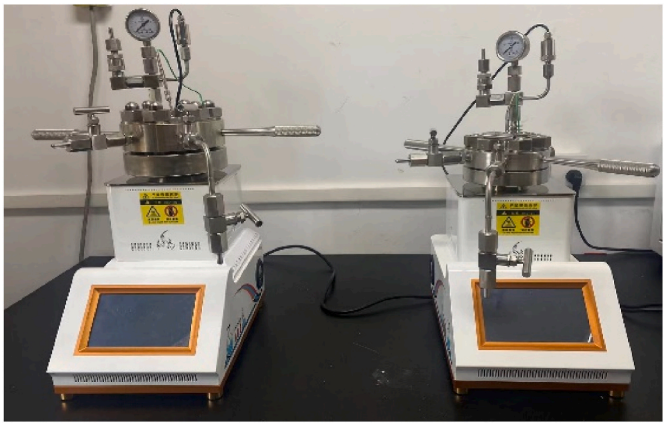


Fig. 1. High-temperature and high-pressure reaction vessels used for CO_2 treatment procedure.

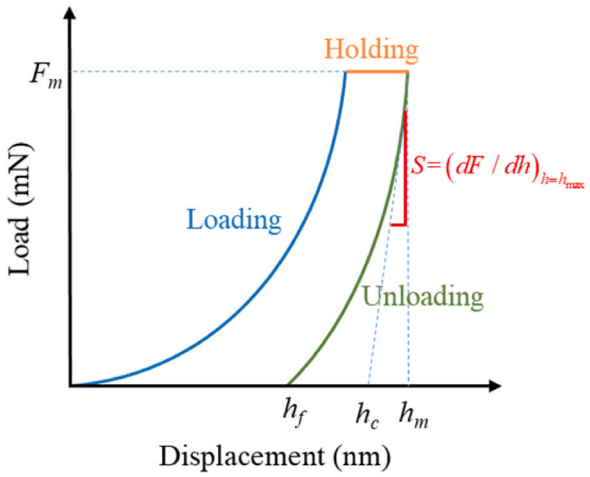


Fig. 2. Typical load-displacement curve (h_f and h_m refer to residual displacement and maximum displacement, respectively).

2.4.2. REA determination and grid nanoindentation design

For heterogeneous materials such as shale, a single or a few indentations cannot provide sufficient information about material properties [37]. A grid nanoindentation method was developed to characterize the mechanical properties of heterogeneous materials [38]. The size of the grid should be larger than the representative elementary area (REA) of the material. The box-counting method was employed to determine the REA of the sample, which is based on the mineral map (obtained from TESCAN Integrated Mineral Analyzer). For detailed information about the box-counting method, please refer to previously published papers [39]. Fig. 3a shows the schematic of the REA determination. The REA of Yanchang shale sample was determined to be 350 μm × 350 μm (Fig. 3b), according to which a 12 × 12 indentation grid was designed. The distance between adjacent indentations is 30 μm, and the area covered by the grid is 360 μm × 360 μm (Fig. 3c).

2.4.3. Statistical deconvolution

Shale is composed of phases with different mechanical properties at the micro level. It is reasonable to make the following assumptions: (1) Due to the sufficiently small detection size, each indentation test measures the mechanical properties of a single phase; (2) The mechanical parameters of each phase obey a Gaussian distribution. Therefore, the overall statistical distribution of the elastic data set measured from the entire sample is a mixture of Gaussian distributions of each phase. Namely, the measured data set follows a Gaussian Mixture Model (GMM). Statistical deconvolution of indentation data is performed to extract the mechanical characteristics of each individual phase. Assuming that there are n mechanical phases in shale, and the mechanical parameters of single phase follow a normal distribution:

$$P_J(x) = \frac{2}{\sqrt{2\pi\left(S_J^x\right)^2}}\exp\left(\frac{-\left(x-\left(U_J^x\right)\right)^2}{2\left(S_J^x\right)^2}\right), x = (E,H) \tag{3}$$

where U_J^x and S_J^x are the mean value and the standard deviation of $x = (E,H)$ of each phase. Based on the assumption that each material phase does not mechanically interact with another phase, the Gaussian mixture model can be used to describe the overall frequency distribution of the mechanical properties, i.e.:

$$P_x = \sum_{J=1}^{n} f_J P_J(x) \tag{4}$$

where P_J is the normal distribution of the material phase and f_J is the

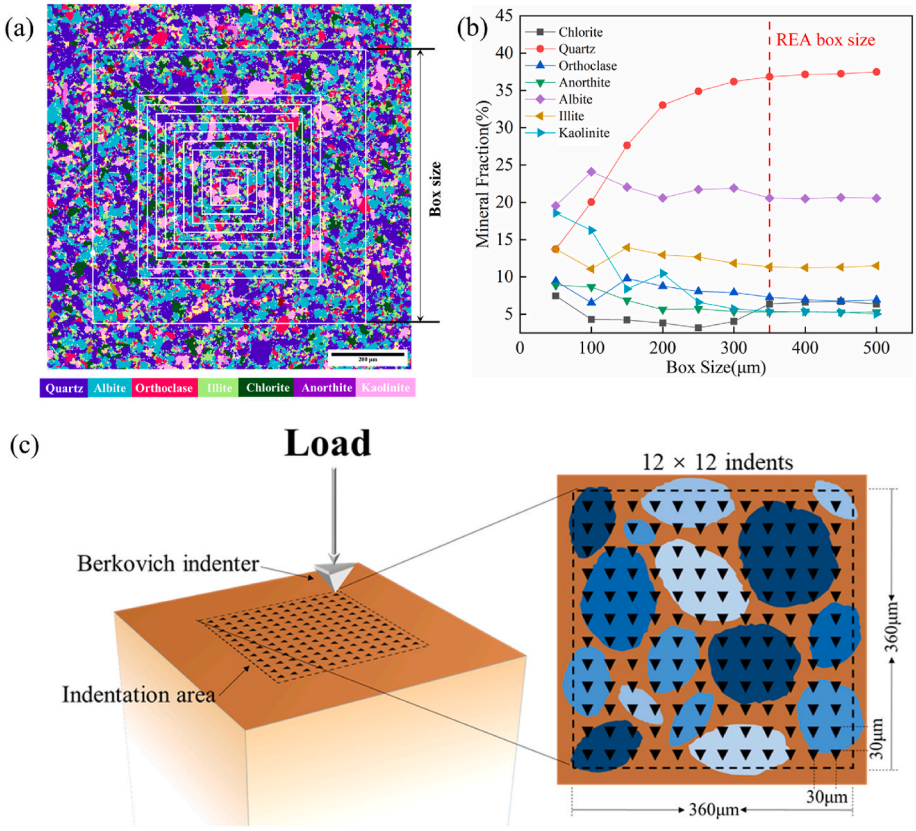


Fig. 3. (a) The schematic of the REA determination. (b) The mineral fraction in different size boxes. (c) The schematic of the nanoindentation design.

volume fraction of the material phase which subjects the constraint:

$$\sum_{J=1}^{n} f_J = 1 \tag{5}$$

The expectation maximization (EM) algorithm is usually used to estimate the mean value, standard deviation, and volume fraction of each phase. There is a detailed introduction to the EM algorithm in previous publications [40].

2.4.4. Mechanical upscaling method

Shale, at the microscopic level, is composed of phases with distinct mechanical properties, and its macroscopic mechanical behavior is determined by the mechanical properties of these individual constituent phases. After obtaining the mechanical properties of individual phases through nanoindentation and statistical deconvolution, the macroscopic mechanical property of the sample can be estimated via an elastic homogenization method. The homogenization method simplifies the complex microstructure into an equivalent continuous medium, the macroscopic mechanical properties of which can be obtained by appropriately averaging the mechanical properties of its individual constituent phases [41]. This approach provides a global, average result that can represent the overall properties of the material. Mori-Tanaka (M-T) method is a weighted averaging scheme that considers a host material with embedded inclusions [42]. It assumes the inclusion (coarse particles) is embedded between layers of the matrix (clay) and then takes into account the interaction of adjacent inclusions. Research conducted by Liu et al. [28] indicates that the mechanical homogenization results derived from the M-T model align closely with the experimental outcomes of uniaxial compression tests, and can serve as a reliable estimation of the macroscale mechanical properties of shale. The equivalent volume modulus K_M and shear modulus G_M of each phase are calculated using the following equations [43]:

$$K_M = \frac{\sum f_J \frac{k_J}{3k_J+4\mu_0}}{\sum \frac{f_J}{3k_J+4\mu_0}} \tag{6}$$

$$G_M = \frac{\sum \frac{f_J \mu_J}{\mu_0(9k_0+8\mu_0)+6\mu_J(k_0+2\mu_0)}}{\sum \frac{f_J}{\mu_0(9k_0+8\mu_0)+6\mu_J(k_0+2\mu_0)}} \tag{7}$$

where, k_0 and μ_0 are the bulk modulus and shear modulus of the reference medium, respectively, while k_J and μ_J are the moduli of the J_{th} phase.

Finally, the elastic properties can be homogenized using the following equation:

$$E_{hom} = \frac{9K_M G_M}{3K_M + G_M} \tag{8}$$

3. Results

3.1. Alteration of mineral content

The mineral contents of untreated sample, CO_2 treated sample, and CO_2+H_2O treated sample were obtained by XRD analysis (Fig. 4). The original mineral composition of shale samples was dominated by quartz and feldspar minerals. The original contents of quartz, albite, and anorthite were 21 wt%, 27 wt%, and 32 wt%, respectively. In addition, there were 9 wt% kaolinite, 4 wt% muscovite, and 6 wt% other minerals in untreated shale sample. The trends of mineral content alteration were the same after CO_2 treatment and CO_2+H_2O treatment, but the alteration was greater after CO_2+H_2O treatment. The contents of quartz and muscovite increased after treatment, while albite, anorthite, and kaolinite decreased. The mineral contents of the sample treated with CO_2 were: quartz 40 wt% (+90.47 %), muscovite 5 wt% (+25 %), albite 20 wt% (−25.93 %), anorthite 25 wt% (−21.88), kaolinite 4 wt%

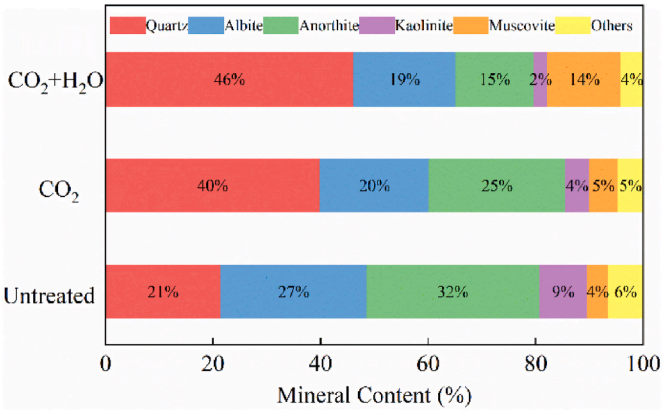


Fig. 4. The mineral contents of untreated sample, CO_2 treated sample, and CO_2+H_2O treated sample were obtained by XRD analysis.

(−55.56 %). The mineral contents of the sample treated with CO_2+H_2O were: quartz 46 wt% (+119.05 %), muscovite 14 wt% (+250.00 %), albite 19 wt% (−29.63 %), anorthite 15 wt% (−53.13 %), kaolinite 2 wt % (−77.78 %). This indicated that the chemical reactions between CO_2 and shale minerals were more intense in the presence of water.

3.2. Alteration of microscale mechanical properties

3.2.1. Load-displacement curves

A few of representative load-displacement curves are demonstrated in Fig. 5. The pattern of load-displacement curve is related to the mechanical properties of the measured component. Generally, soft component yields large displacements, while stiff component yields small displacements. After CO_2 and CO_2+H_2O treatment, the maximum displacement and residual displacement of the soft component increased, indicating that the soft component became softer and its plasticity increased. In addition, pop-in phenomenon (the sudden jump in the load–displacement curve) occurred. Pop-in was caused by grain dislocations, pore collapse, or crack propagation beneath the indenter [44–46]. This indicated that after CO_2/CO_2+H_2O treatment, additional pores generated in the shale, and the mechanical performance of minerals altered.

3.2.2. Elastic modulus and hardness

The arrangement of indentation locations on the sample surface are shown in Fig. 6a–c. Based on the mechanical parameters measured at each indentation location, the two-dimensional distributions of elastic modulus (E) and hardness (H) were obtained through linear interpolation method (Fig. 6d–i). It can be found that the spatial distribution of E and H at the microscale is heterogeneous, which originated from the heterogeneity of shale composition. After CO_2 treatment, the proportion of high elastic modulus and hardness regions decreased, while that of low elastic modulus and hardness regions increased. This phenomenon was more obvious in the CO_2+H_2O treated shale.

3.3. Deconvolution results

Fig. 7 illustrates the deconvolution results of untreated sample, CO_2 treated sample, and CO_2+H_2O treated sample. Three mechanical phases (i.e. soft phase, intermediate phase, and stiff phase) were deconvolved from the nanoindentation results. The macroscopic mechanical behavior of shale is jointly determined by the mechanical phases contained. Exploring the changes in each mechanical phase after CO_2 treatment helps to fundamentally understand the softening behavior of shale. As shown in Fig. 8, in regard to the untreated sample, the mean values of E for soft phase, intermediate phase, and stiff phase are 60.45 GPa, 80.86 GPa, and 218.87 GPa, respectively. Concerning CO_2 treated sample, the mean values of E for soft phase, intermediate phase, and stiff phase are 53.31 GPa, 82.93 GPa, and 133.77 GPa, respectively. In regard to CO_2+H_2O treated sample, the mean values of E for soft phase, intermediate phase, and stiff phase are 29.30 GPa, 49.33 GPa, and 80.98 GPa, respectively. It can be observed that under CO_2 treatment condition, both the hard phase and the soft phase softened, while the intermediate phase did not. Under the CO_2+H_2O treatment condition, all three phases - hard, intermediate, and soft - experienced softening, and the softening of each phase was more significant compared to that in the absence of water. The results of deconvolution are concluded in Table 1.

3.4. Upscaling of mechanical properties

Homogenization is a crucial step when upscaling the mechanical properties from the microscale to a larger scale level. The fraction of each phase, obtained by deconvolution, and the volume fraction are identical in perfectly disordered material [47]. In this study, the results obtained by deconvolution were used as input to the M-T model to obtain the homogenized elastic modulus, which was proven feasible in previously published literature [28]. As shown in Fig. 9, the homogenized E of untreated sample, CO_2 treated sample, and CO_2+H_2O treated sample are 62.08 GPa, 55.30 GPa, and 33.67 GPa, respectively. Compared with untreated sample, the decrease in macroscale elastic modulus of CO_2 treated sample is 10.92 %, and that of CO_2+H_2O treated sample is 45.76 %. The deconvolution results suggest that under the condition of CO_2 treatment, the macroscale softening of shale is contributed by the hard and soft phases. While under the condition of CO_2+H_2O treatment, all three phases - hard, intermediate, and soft - contributed to this softening. Consequently, this softening induced by CO_2 is more pronounced in the presence of water. It can be seen from the Table 1 that the fraction of soft phases in untreated, CO_2-treated,

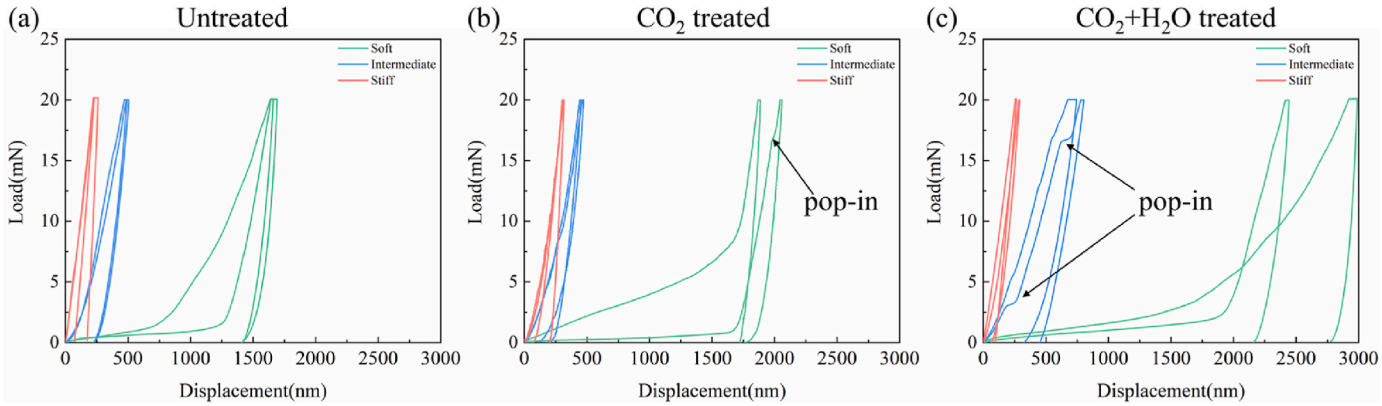


Fig. 5. The load-displacement curves of: (a) Untreated shale sample. (b) CO_2 treated shale sample. (c) CO_2+H_2O treated shale sample.

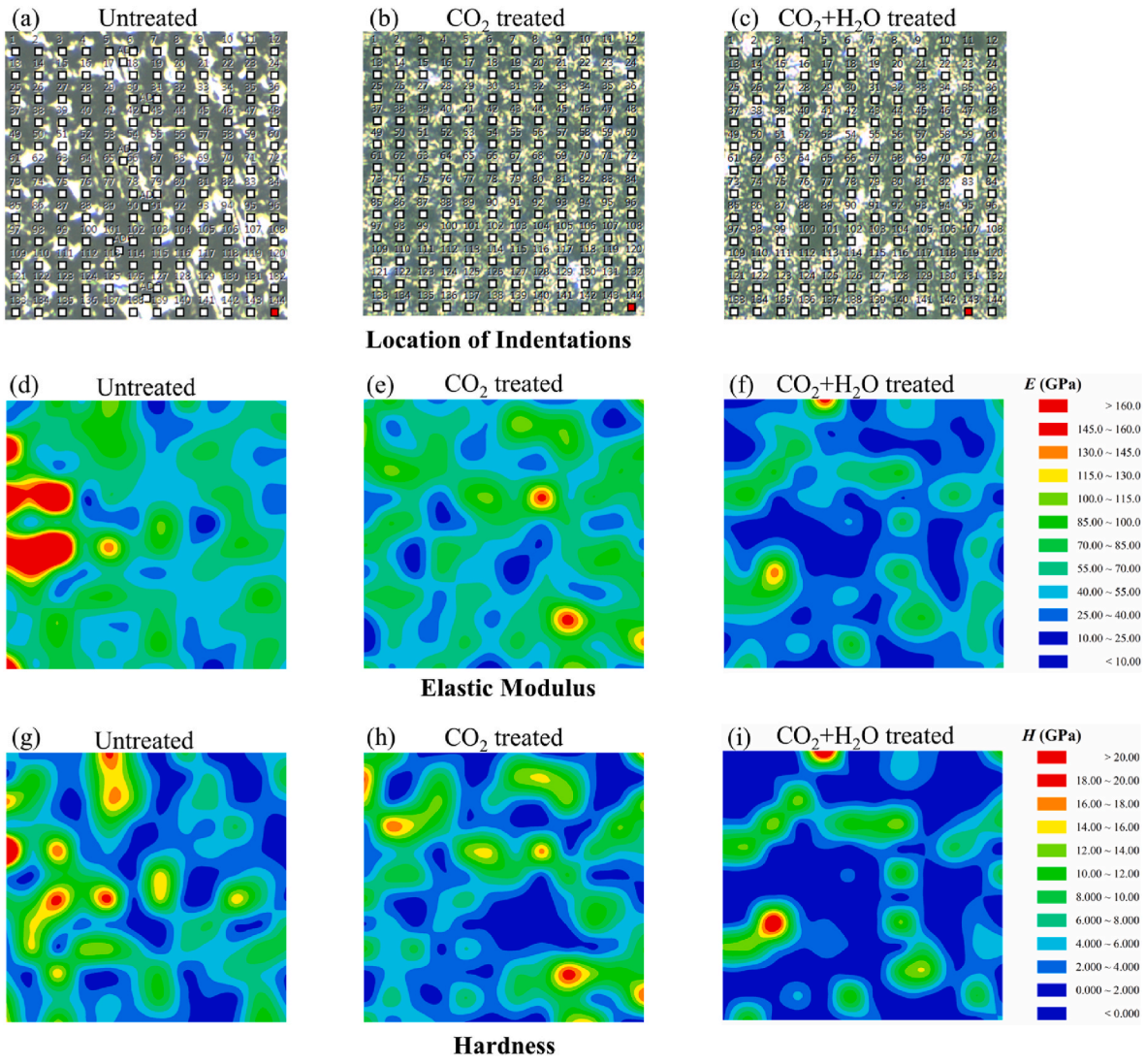


Fig. 6. The arrangement of indentation locations of: (a) Untreated sample, (b) CO_2 treated sample, (c) CO_2+H_2O treated sample. Two-dimensional distributions of elastic modulus (*E*) of: (d) Untreated sample, (e) CO_2 treated sample, (f) CO_2+H_2O treated sample. Two-dimensional distributions of hardness (*H*) of: (g) Untreated sample, (h) CO_2 treated sample, (i) CO_2+H_2O treated sample.

CO_2+H_2O treated samples all exceed 60 %. This suggests that the macromechanical performance of Yanchang shale is mainly controlled by the soft phase. The significant reduction in the macroscale elastic modulus of shale after CO_2+H_2O treatment is primarily due to the substantial decrease in the elastic modulus of the soft phase.

3.5. Comparison with prior studies

We reviewed recently published studies related to this paper, involving shale from different geological formations and various treatment conditions (Table 2). The uniaxial compression test is a common method to characterize shale softening at the macroscale. Zhou et al. [23] conducted CO_2/CO_2+H_2O soaking and uniaxial compression tests on Wufeng shale, and found that after soaking in Sc-CO_2 and Sc-CO_2+H_2O, the elastic modulus decreased by 10.03 % and 12.38 % respectively. Furthermore, they conducted similar tests on Longmaxi shale treated with Sc-CO_2/brine/Sc-CO_2+brine. It was found that the elastic modulus decreased by 4.89 %, 15.77 %, and 19.63 % respectively after treatment with Sc-CO_2, brine, and Sc-CO_2+brine [24].

Other researchers employed nanoindentation to quantify the softening of shale constituents after exposure to different fluids. Ozotta et al. [32] found that the hard phase in Bakken shale underwent significant softening after CO_2 treatment, but not every phase softened. The homogenized elastic modulus obtained through the M-T model decreased with increasing treatment time. It's similar to the observation of our study. Treatment with water also results in the softening of shale. Liu et al. [29] reported that after exposure to moisture for 20 days, the average indentation modulus of Marcellus shale could decrease by up to 15 %. Li et al. [33] investigated the softening behavior of Shahejie shale induced by water/CO_2-rock interaction. Statistical results from the nanoindentation tests indicated that the elastic modulus decreased by 30.0 % after soaking in CO_2 for 6 days, and by 57.6 % after soaking in water for the same period. Cheng et al. [31] obtained changes in the elastic modulus of different minerals in Longmaxi shale after 60-d CO_2+H_2O treatment using nanoindentation and SEM. They found that The values in specific shale minerals are also declined except for the quartz because of its strong mechanical strength and chemical stability. The decreases of elastic modulus are 5.18 %, 24.45 % and 38.23 % for calcite, dolomite, and clay minerals, respectively.

From the above studies, a pattern consistent with this research can be derived, that is, the softening of shale caused by CO_2+H_2O treatment is more pronounced than that caused by CO_2 treatment alone. At the microscale, different constituents in shale exhibit different softening behaviors. The variations in softening degree observed in different

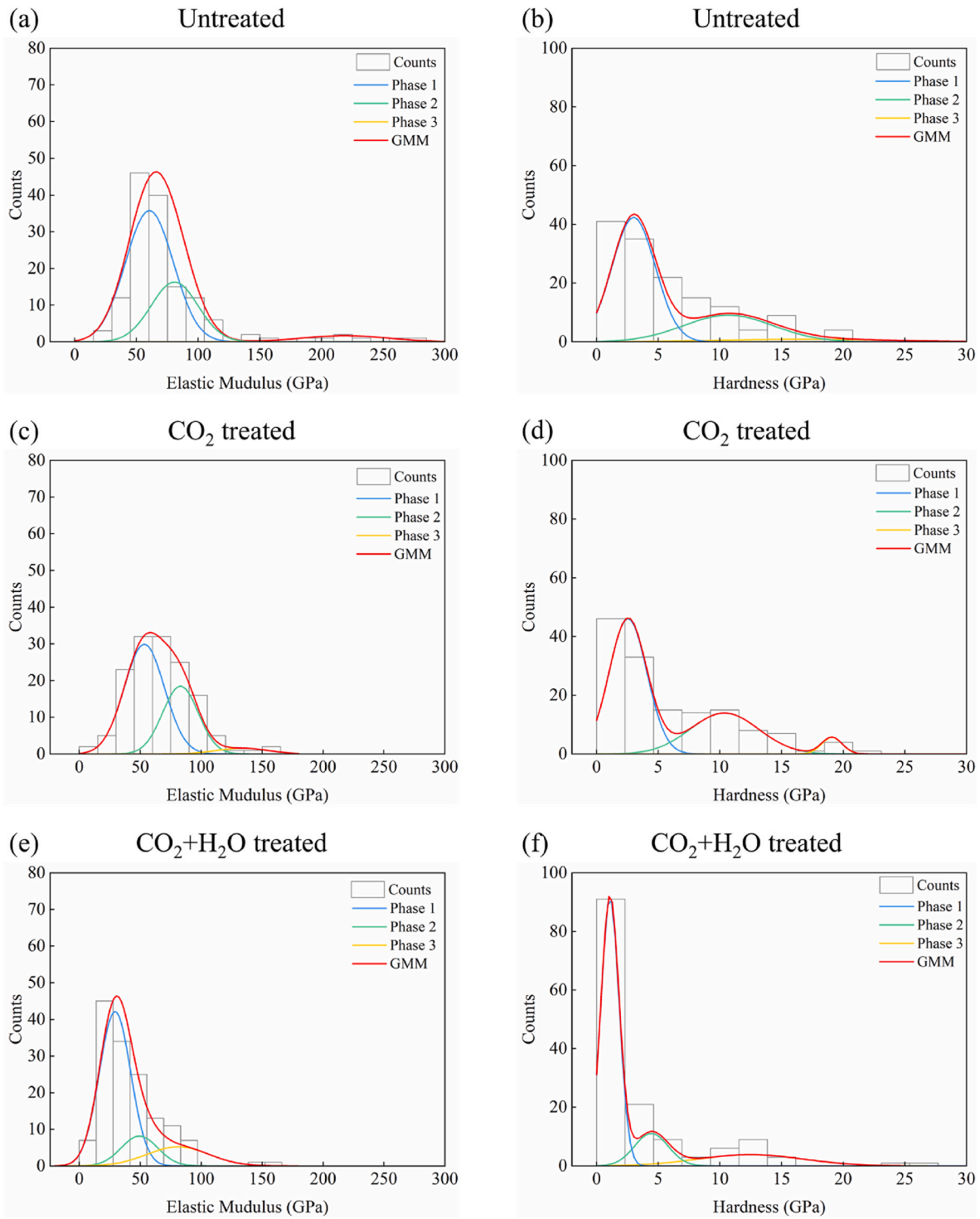


Fig. 7. The deconvolution results: (a) *E* of untreated sample. (b) *H* of untreated sample. (c) *E* of CO_2 treated sample. (d) *H* of CO_2 treated sample. (e) *E* of CO_2+H_2O treated sample. (f) *H* of CO_2+H_2O treated sample.

studies are due to differences in the samples and treatment conditions.

4. Discussion

4.1. Softening mechanism of shale induced by CO_2 treatment

4.1.1. In the absence of water

CO_2 is in a supercritical state at 50 °C and 12 MPa, which has been evidenced to affect the structure of shale. The heterogeneous expansion caused by CO_2 adsorption and mineral dissolution are considered dominant factors. CO_2 adsorption can lead to a decrease in surface energy of shale, leading to volumetric expansion. It will cause the structure of the shale to become loose and soften. Mineral dissolution is another cause of shale softening. The alteration of mineral contents was detected by XRD analysis after CO_2 treatment, which showed that the proportion of albite, anorthite and kaolinite decreased. It should be mentioned that mineral dissolution is mostly an ion reaction that occurs in the presence of water. Although no water was added in the CO_2 treatment

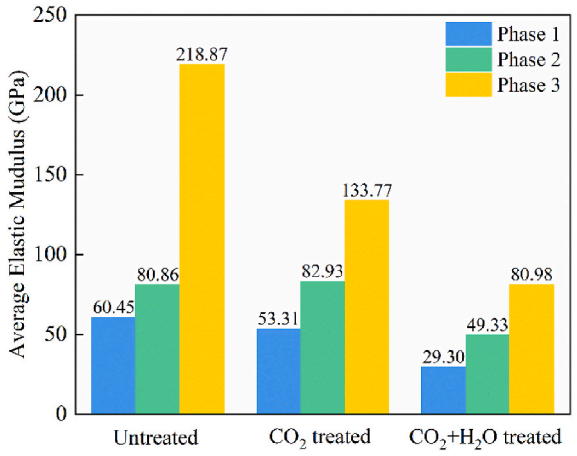


Fig. 8. The average elastic modulus of each phase.

experiment, the bound water originally contained in shale made it possible for ion reactions to occur. It is worth noting that the intermediate phase of the CO_2 treated sample did not soften, but remained at a similar level to the untreated sample. This observation might be ascribed to the negligible influence exerted on the mechanical properties of certain structurally stable minerals under dry CO_2 treatment. However, the specific minerals correspond to each indentation test cannot be obtained through statistical deconvolution. This is a limitation of the research method used in this study.

4.1.2. *In the presence of water*

In the presence of water, the mechanism of softening of shale will differ. CO_2 adsorption induced expansion did not occur in water saturated shale, but was replaced by water absorption induced expansion, which was also a vital factor of shale softening. Additionally, mineral dissolution reactions were more intense in the presence of water. Therefore, the softening behavior of CO_2+H_2O treated shale was dominated by water absorption induced expansion and more intense dissolution reactions.

The softening of shale in the presence of water was significantly greater than that in the absence of water. It might be due to the different extent of dissolution reactions in these two cases. In CO_2 treated sample, dissolution reactions occurred only in localized areas where bound water was present. Moreover, in a limited, disconnected solution volume, the ionic reaction process was self-limited by the increasing concentration of reaction products [48]. In CO_2+H_2O treated sample, dissolution reactions occurred uniformly in the shale. Meanwhile, the reaction products could be transported to other low concentration regions by diffusion, avoiding self-limitation due to the aggregation of reaction products.

4.2. *Implication of shale gas development and CO_2 geological storage*

The softening of shale will have a series of adverse effects on the development of shale gas, such as reduced crack openings, reservoir subsidence, wellbore instability, and proppant embedment [17]. The results of this study show that shale softening induced by CO_2 injection is more severe in the presence of water than that in the absence of water. The inspiration is that when selecting the target area for CO_2 fracturing, a highly water-saturated reservoir should be avoided as much as possible. In addition, due to the weak proppant carrying ability of CO_2, proppants are usually mixed with water-based fracturing fluids and injected into the reservoir after CO_2 fracturing. The reverse discharging of water-based fracturing fluid should be promoted as soon as possible, or a more optimized proppant injection strategy should be adopted to avoid long-term exposure of shale to CO_2+H_2O environment.

Additionally, shale is a common caprock type for CO_2 geological sequestration sites [49]. When serving as a caprock, shale is typically water saturated [50,51]. This indicates that the caprock will undergo significant softening behavior after CO_2 injection. When designing CO_2 geological sequestration injection strategies, the mechanical properties of the caprock are a critical input parameter. The softening effect induced by CO_2 injection should be considered to avoid overestimating the CO_2 sequestration capacity and the consequent potential safety hazards.

5. Conclusion

In this study, the softening behavior of shale induced by CO_2 injection was investigated using nanoindentation. Emphasis was placed on analyzing the differences in shale softening behavior in the absence and presence of water. The main conclusions are as follows:

(1) CO_2 injection caused alteration of shale mineral composition. The trends of mineral content alteration were the same in both CO_2 treatment and CO_2+H_2O treatment conditions, but the alteration was greater in the presence of water.
(2) Three mechanical phases (i.e. soft phase, intermediate phase, and stiff phase) were deconvolved from the nanoindentation results. under CO_2 treatment condition, both the hard phase and the soft phase softened, while the intermediate phase did not. Under the CO_2+H_2O treatment condition, all three phases - hard, intermediate, and soft - experienced softening, and the softening of each phase was more significant compared to that in the absence of water.

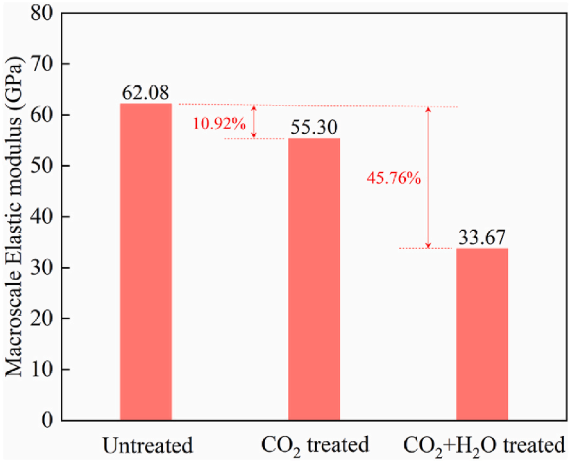


Fig. 9. The macroscale elastic modulus obtained through M-T model.

Table 1
The results of deconvolution. (μ = mean value, σ = standard deviation, f = fraction).

Sample	Phase 1			Phase 2			Phase 3		
	μ, GPa	σ, GPa	f, %	μ, GPa	σ, GPa	f, %	μ, GPa	σ, GPa	f, %
Untreated	60.45	19.41	65.28	80.86	19.01	29.17	218.87	35.24	5.55
CO_2 treated	53.31	16.50	61.81	82.93	14.69	34.03	133.77	21.08	4.16
CO_2+H_2O treated	29.30	12.88	68.05	49.33	14.95	15.28	80.98	25.45	16.67

Table 2
Prior studies related to this paper.

Ref.	Shale Formations	Mechanical method	Treatment condition	Principal Conclusions
[23]	Wufeng	Uniaxial compression	CO_2/CO_2+H_2O (14 MPa, 45 °C, 15 d)	After soaking in Sc-CO_2 and Sc-CO_2+H_2O, the elastic modulus decreased by 10.03 % and 12.38 % respectively.
[24]	Longmaxi	Uniaxial compression	CO_2/brine/CO_2+brine (12 MPa, 45 °C, 10d)	The elastic modulus decreased by 4.89 %, 15.77 %, and 19.63 % respectively after treatment with Sc-CO_2, brine, and Sc-CO_2+brine.
[32]	Bakken	Nanoindentation	CO_2 (7.4 MPa, 31 °C, 3d, 8d, 16d, 30d, 60d)	The hard phase underwent significant softening, but not every phase softened. The homogenized elastic modulus obtained through the M-T model decreased with increasing treatment time.
[29]	Marcellus	Nanoindentation	H_2O (20d)	The average indentation modulus of could decrease by up to 15 %.
[33]	Shahejie	Nanoindentation	CO_2/H_2O (20 MPa, 80 °C, 2d, 6d)	The elastic modulus decreased by 30.0 % after soaking in CO_2 for 6 days, and by 57.6 % after soaking in water for the same period.
[31]	Longmaxi	Nanoindentation	CO_2+H_2O (13.8 MPa, 200 °C, 90d)	The decreases of elastic modulus are 5.18 %, 24.45 % and 38.23 % for calcite, dolomite, and clay minerals, respectively.

(3) The M-T model was employed to estimate the macroscale elastic modulus of shale samples. Compared with untreated sample, the decrease in the macroscale elastic modulus of CO_2 treated sample is 10.92 %, and that of CO_2+H_2O treated sample is 45.76 %.
(4) To mitigate the adverse effects caused by shale softening, it is advisable to avoid targeting reservoirs with high water saturation during CO_2 fracturing. Simultaneously, efforts should be made to expedite the backflow of water-based fracturing fluids used for proppant transport. Moreover, when designing the CO_2 geological sequestration injection strategy, the softening effect induced by CO_2 injection should be considered to prevent overestimating the CO_2 sequestration capacity and the consequent potential safety risks.

Credit author statement

Qi Cheng: Investigation, Writing – Original Draft, Writing – Review & Editing; **Jiren Tang:** Conceptualization, Supervision, Funding acquisition; **Yunzhong Jia:** Writing – Review & Editing; **Yiyu Lu:** Conceptualization, Funding acquisition; **Chi Zhang:** Investigation; **Yanlin Liu:** Investigation; **Guilin Zhao**: Investigation; **Yalu Liu:** Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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